RAJJWOL SINGH MAHARJAN

Backend & Software Engineer | Full-Stack Developer

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| GitHub: github.com/Rajjwol | Portfolio: rajjwol.github.io/my-portfolio/

PROFESSIONAL SUMMARY

Full-Stack & Backend Engineer experienced in building end-to-end web and mobile applications using C# (.NET Core), PHP (Laravel), React (TypeScript), and Kotlin. Demonstrated track record in designing multi-criteria recommendation engines, RESTful APIs, and responsive Ul interfaces. Skilled in MongoDB/MySQL schema design, state management, and modern software engineering best practices.

TECHNICAL SKILL

- **Programming Languages:** C# (.NET Core), PHP, Python, JavaScript, TypeScript, Java, Kotlin, SQL
- **Backend & Frameworks:** ASP.NET Core Web API, Laravel, React.js, Jetpack Compose, RESTful APIs
- **Databases & Systems:** MongoDB, MySQL, Linux (Ubuntu/Debian), Windows
- **Tools & Security:** Git, VS Code, Postman, Swagger/OpenAPI, JWT, Google OAuth 2.0, MailKit, BCrypt
- **Soft Skills:** Teamwork, Problem-solving, Communication skill

KEY TECHINICAL PROJECT

MeroDiet: AI-Based Personalized Health & Fitness System

Technologies: Kotlin | Android Jetpack Compose | ASP.NET Core & MongoDB | Retrofit | DataStore

- Architected an AI-driven health platform connecting a native Kotlin/Jetpack Compose Android app to an ASP.NET Core & MongoDB backend engine.
- Developed an Android client using MVVM, Clean Architecture, StateFlow, and Coroutines for seamless UI state management and offline capability.
- Implemented a personalization engine utilizing MAUT ranking, collaborative filtering, and dynamic user feedback loops (Accept/Reject/Ban).
- Integrated Retrofit REST services, DataStore local persistence, Coil proxy image caching, and ExoPlayer for workout video streaming.

AIPlanner: Smart Nutrition & Meal Recommendation Backend Engine

Technologies: C# | ASP.NET Core Web API | MongoDB | JWT & Google OAuth 2.0 | Swagger | MailKit

- Engineered an ASP.NET Core REST API backed by MongoDB, exposing 15+ endpoints for personalized food scoring and weekly meal generation.
- Built a rule-based Safety Filter Algorithm enforcing allergen exclusions and medical condition threshold rules (e.g., sodium and sugar caps).
- Implemented JWT Bearer authentication, refresh token rotation, Google OAuth 2.0 account merging, and MailKit OTP verification.
- Developed WeeklyMealPlanAlgorithm for 7-day meal schedule generation and automated database initialization using DbSeeder.

Oceanic Delights: Fish Recommendation System

Technologies: PHP | MySQL | JavaScript | HTML5 / CSS3

- Architected a full-stack web application for aquarium e-commerce using PHP and MySQL, incorporating secure cart management, user authentication, and product cataloging.
- Engineered a hybrid recommendation engine combining Content-Based and Collaborative Filtering, utilizing Cosine Similarity to score vector similarities across user cart items and purchase history.
- Optimized MySQL database schema design and indexing to achieve sub-second query execution for real-time recommendation generation and dynamic catalog search.

Gender Classification System (Machine Learning Project)

Technologies: Python | Scikit-learn | Pandas | NumPy

- Built a machine learning pipeline in Python using scikit-learn, pandas, and numpy for demographic data preprocessing and feature extraction.
- Trained and benchmarked classification algorithms including K-Nearest Neighbors (KNN) and Decision Trees to optimize prediction accuracy.

WORK EXPERIENCE

Full-Stack Software Engineering Intern | **Phye Gan Pvt. Ltd.(Photographer Diary)**

Kathmandu, Nepal • 3 Months Internship

- Architected a full-stack web application using Laravel (PHP) and React (TypeScript) with RESTful API integration for secure, scalable communication.
- Engineered core system features including user authentication, album and photo management, profile customization, and optimized database queries.
- Designed a responsive UI/UX using React and Tailwind CSS, ensuring cross-device compatibility, clean state management, and optimized asset loading.

EDUCATION

Bachelor of Computer Applications (BCA) | National College of Computer Studies, Kathmandu/Paknajol
(Expected Graduation: 2026)
+2 in Management (Computer Science) | New Zenith English Model School, Kathmandu/Dekwa Galli